

Midterm 2 will cover topics from 3.1, 3.2, 3.3, 3.4, 3.7, 2.1 (and some in 1.2), and 2.2

- Problem 1 - Set up the appropriate form of partial fraction decomposition for a rational function. (Section 3.4)
- Problem 2 - Complete the arguments of using comparison test to determine if an improper integral is convergent or divergent. (Section 3.7)
- Problem 3 - Integrals using u -substitution, integration by parts, partial fractions or trigonometric substitutions. (Section 3.1, 3.2, 3.3, 3.4)
- Problem 4 - Evaluate improper integral using limit notation. (Section 3.7)
- Problem 5 and 6 - Find average value of a continuous function on a closed interval. (Section 1.2)
- Problem 7, 8, 9, 10 and 11 - Find the area between two curves. (Section 2.1)
- Problem 11, 12, 13 and 14 - Find the volume of a solid of revolution by disk/washer method. (Section 2.2)

Formula Sheet

$$\sin^2 x + \cos^2 x = 1, \quad \sin^2 x = 1 - \cos^2 x, \quad \cos^2 x = 1 - \sin^2 x$$

$$\sec^2 x = \tan^2 x + 1, \quad \tan^2 x = \sec^2 x - 1$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}, \quad \sin 2x = 2 \sin x \cos x$$

$$\sin A \cos B = \frac{1}{2}[\sin(A - B) + \sin(A + B)]$$

$$\cos A \cos B = \frac{1}{2}[\cos(A + B) + \cos(A - B)]$$

$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$\int \cot x \csc x \, dx = -\csc x + C, \quad \int \csc^2 x \, dx = -\cot x + C$$

$$\int \tan x \, dx = \ln |\sec x| + C, \quad \int \cot x \, dx = \ln |\sin x| + C$$

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C, \quad \int \csc x \, dx = \ln |\csc x - \cot x| + C$$

$$\int \sec^3 x \, dx = \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln |\sec x + \tan x| + C$$

$$\int \frac{1}{1+x^2} \, dx = \arctan x + C, \quad \int \frac{1}{\sqrt{1-x^2}} \, dx = \arcsin x + C$$

$$\int \frac{1}{a^2+x^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C, \quad \int \frac{1}{\sqrt{a^2-x^2}} = \arcsin\left(\frac{x}{a}\right) + C$$

$$\int_1^\infty \frac{1}{x^p} \, dx = \begin{cases} \text{is convergent} & \text{if } p > 1 \\ \text{is divergent} & \text{if } 0 < p \leq 1. \end{cases} \quad \int_0^1 \frac{1}{x^p} \, dx = \begin{cases} \text{is convergent} & \text{if } 0 < p < 1 \\ \text{is divergent} & \text{if } p \geq 1. \end{cases}$$

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined

1. Set up the partial fraction decomposition of the following rational function. DO NOT determine the numerical values of the coefficients. DO NOT integrate the function.

1). $\frac{2 - x + x^2 + 4x^3}{(x + 4)^3(x^2 + 4)} =$ _____

2). $\frac{x^2 - 2x + 5}{(x^2 + 9)x^2(x - 1)} =$ _____

3). $\frac{x^3 + 2x - 1}{(x^2 + 1)^2(x + 2)} =$ _____

2. For each of the following improper integral, one may use Comparison Test to determine if they are convergent or divergent. Circle the correct answers to complete the arguments of using the comparison test.

- 1). The improper integral $\int_1^{\infty} \frac{1}{\sqrt[7]{x^4 - 0.3}} dx$ is Convergent or Divergent (circle the correct answer) because $\frac{1}{\sqrt[7]{x^4 - 0.3}}$ _____ (choose one between A and B, and then choose one among C, D, E, and F).

A). $\geq$ B). $\leq$, C). $\frac{1}{x^4}$, D). $\frac{1}{x^7}$, E). $\frac{1}{x^{7/4}}$, F). $\frac{1}{x^{4/7}}$

- 2). The improper integral $\int_1^{\infty} \frac{x^{3/2}}{\sqrt{x^6 + 3}} dx$ is Convergent or Divergent (circle the correct answer) because $\frac{x^{3/2}}{\sqrt{x^6 + 3}}$ _____ (choose one between A and B, and then choose one among C, D, E, and F).

A). $\geq$ B). $\leq$, C). $\frac{1}{x}$, D). $\frac{1}{x^{3/2}}$, E). $\frac{1}{x^{1/6}}$, F). $\frac{1}{x^6}$

- 3). The improper integral $\int_1^{\infty} \frac{\sin^2 x}{\sqrt[3]{x^5 + 3}} dx$ is Convergent or Divergent (circle the correct answer) because $\frac{\sin^2 x}{\sqrt[3]{x^5 + 3}}$ _____ (choose one between A and B, and then choose one among C, D, E, and F).

A). $\geq$ B). $\leq$, C). $\frac{1}{x^3}$, D). $\frac{1}{x^5}$, E). $\frac{1}{x^{3/5}}$, F). $\frac{1}{x^{5/3}}$

3. Evaluate the following integrals.

1). $\int_0^{\pi/3} \sin^2 \theta \cos^3 \theta d\theta$

2). $\int \tan^3 \theta \sec^4 \theta d\theta$

3). $\int x e^{-2x} dx$

4). $\int \frac{\ln x}{\sqrt{x}} dx$

5). $\int x^2 \ln x dx$

6). $\int \sin^2 x \tan x dx$

7). $\int \frac{\sqrt{x^2 - 4}}{x^2} dx$

8). $\int \frac{x^2}{\sqrt{9 - x^2}} dx$

9). $\int \frac{x^2}{(1 + x^2)^{3/2}} dx$

10). $\int \frac{1}{x^3 \sqrt{x^2 - 1}} dx$

11). $\int_0^1 \frac{x^2}{(4 - x^2)^{3/2}} dx$

12). $\int \frac{3x + 6}{x^2 - 3x} dx$

13). $\int \frac{4x + 2}{x^2 - 2x - 8} dx$

14). $\int \frac{4x^2 - x + 8}{x(x^2 + 4)} dx$

4. Evaluate the following improper integrals or show that they are divergent.

1). $\int_1^{\infty} \frac{(\tan^{-1} x)^2}{x^2 + 1} dx$

2). $\int_1^4 \frac{1}{(x - 1)^{2/3}} dx$

3). $\int_1^{\infty} \frac{3}{(x + 1)^3} dx$

4). $\int_{-2}^0 \frac{1}{(x + 2)^2} dx$

5. Find the average value of the function $f(x) = x^2 \sqrt{1 + x^3}$ on $[0, 2]$.

6. In a certain city the temperature (in degrees Fahrenheit) t hours after 9am was approximated by the function

$$T(t) = 70 + 20 \sin\left(\frac{\pi t}{12}\right).$$

Find the average temperature during the period from 9am to 9pm.

7. Sketch the graph of the region bounded by $y = x + 2$ and $y = 8 - x^2$. Then find the area of the region.

8. Sketch the graph of the region bounded by $y = \sqrt{x}$ and $y = \frac{1}{2}x$. Then find the area of the region.

9. Sketch the graph the region bounded by $y = \sin x$, $y = e^x$, $x = 0$ and $x = \frac{\pi}{2}$. Then find the area of the region.

10. Sketch the graph of the region bounded by $x = 1 - y^2$ and $x = y^2 - 1$. Then find the area of the region.
11. Sketch the region enclosed by the curves $y = x + 1$, $y = 7 - x^2$, $x = 0$ and $x = 4$. Then find the area of the region.
12. Use disk/washer method, find the volume of the solid obtained by rotating the region bounded by $y = 1 - x^2$ and $y = 0$ about x -axis. Make sure to sketch the graph of the given region.
13. Use disk/washer method, find the volume of the solid obtained by rotating the region bounded by $y = x$, $y = \sqrt{x}$ about x -axis. Make sure to sketch the graph of the given region.
14. Use disk/washer method, find the volume of the solid obtained by rotating the region bounded by $x = 2\sqrt{y}$, $x = 0$, and $y = 9$ about y -axis. Make sure to sketch the graph of the given region.
15. Use disk/washer method, find the volume of the solid obtained by rotating the region bounded by $y = 5 - x^2$ and $y = \frac{1}{4}x^2$, about x -axis. Make sure to sketch the graph of the given region.